

NT-ops: Number Theory Algorithms and Optimisations

Pre-requisites

- **Pre-requisites:** Basic C/C++ and fundamental programming concepts (ESC111/112).

Logistics

- **Classes:** 3 classes per week (mostly nights).
- **Platform:** Google Meet (link will be shared in the group).
- **Practice Sessions:** Mentors will solve problems and do live coding.
- **Contest Solving Sessions:** Mentors will solve problems from the mid-eval and end-eval contests.
- **Resources:** All lecture slides, assignments, solutions, and reference materials will be available via a centralized Drive link.

Expected Engagement & Certification

- **Attendance:** Attendance is taken in every class.
- **Evaluations:** Mid-eval and end-eval contests are mandatory.
- **Ratification:** Only the top 80% of participants who complete the evaluations will be ratified.

Assignments

- Assignments will be shared after every class, containing practice questions.
- **Submission:** Only the starred (★) questions need to be submitted via Google Forms.

Course Roadmap

- **Week 1: Fundamentals & Linear Diophantine Equations**
 - Introduction & Basics
 - Divisibility & Euclidean Algorithms
 - Linear Diophantine Equations
- **Week 2: Primes & Congruences**
 - Primes & Sieves
 - Advanced Modular Arithmetic
 - System of Congruences
- **Week 3: Sequences & Combinatorics**

- Fibonacci Numbers
- Common Number-Theoretic Functions
- Modular Combinatorics
- **Week 4: Mid-term Evaluation & Convolutions**
 - Mid-Evaluation Contest
 - Mid-Evaluation Upsolving
 - Dirichlet Convolutions
- **Week 5: Inversion & Advanced Primality**
 - Mobius Inversion Formula
 - Fast Primality Checking
 - Integer Factorisation
- **Week 6: Advanced Algorithms & Final Eval**
 - Quadratic Residues
 - Modular Square Roots
 - End evaluation contest

Mentors

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